Physical Computing

Embedded Programming

* Min res 56x16 char

Material

- Microcontroller board,
  e.g. Feather M4 Express
- USB C to micro cable
- Half-sized breadboard

Getting started

– Install Arduino IDE
  Version 2.?.?

* https://www.arduino.cc/en/software/#ide

Set up the board

- Preferences > Additional Boards Manager URL
  https://adafruit.github.io/arduino-board-index/package_adafruit_index.json
- Tools > Board > Boards Manager... > Adafruit SAMD Boards
- Tools > Board > Adafruit M4 Feather Express
- Tools > Port > ...

* https://github.com/tamberg/fhnw-phycom/wiki/Feather-M4-Express-Board

Microcontroller

- Small computer
- Runs one program
- No operating system

GPIO Pins

- Input or output
- Digital (1/0) or
  analog (1 .. 1024)
- M4 has 3.3V logic
- Check the pinout

* https://learn.adafruit.com/adafruit-feather-m4-express-atsamd51/pinouts

Blinking an LED

- Arduino IDE > Examples > Basics > Blink
- Search "Arduino LED circuit"
- Build circuit on breadboard, use 1k resistor
- Use on-board USB power
- Adapt the pin number

```
# Printing "hello"

void setup() {
  Serial.begin(115200);
}

void loop() {
    Serial.println("hello");
    delay(500);
}
```

Monitoring output

- Arduino > Tools > Serial Monitor
- Plug in the M4 boad via USB
- Select the baud rate 115200
- Or whatever is used in code

Reading a button

- Arduino IDE > Examples >
  Basics > DigitalReadSerial
- Search "Arduino Button circuit"
- Build circuit on breadboard,
  use 10k resistor
- Use on-board USB power
- Adapt the pin number

Making a dial

- Arduino IDE > Examples > Basics > AnalogReadSerial
- Search "Arduino Poti circuit"
- Build circuit on breadboard, use 10k resistor
- Adapt the pin number
- Use the map() function